

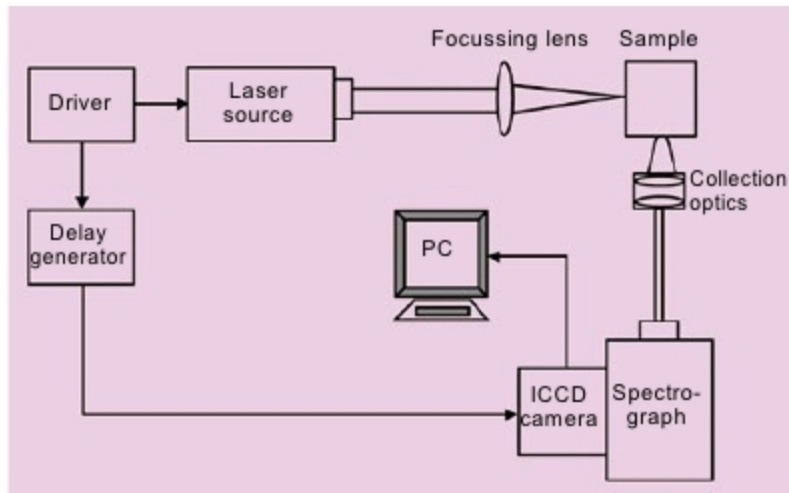


Fig. 9: Laser-induced breakdown spectroscopy setup



Fig. 10: Field testing of LIBS explosive detector military prototype of Applied Spectra

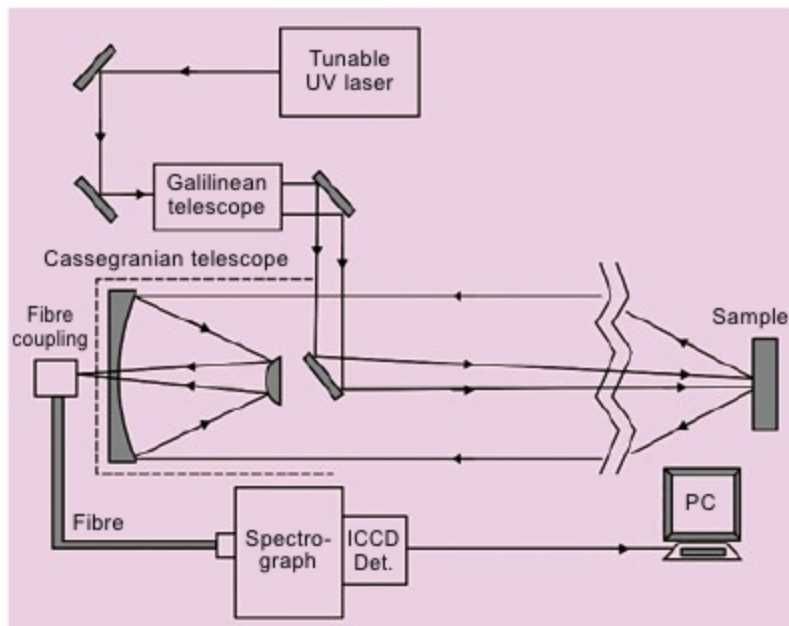


Fig. 11: Typical Raman spectroscopy setup

reception, photomultiplier tube (PMT) for recording backscattered and bio-fluorescence signals, spectrograph with gated ICCD array for recording dispersed fluorescence spectra and lidar controller.

Fourth harmonic at 266nm is transmitted towards the biological cloud. The telescope receives backscattered and bio-fluorescence signals, which are then fed to PMT channel. Back-scattered signal is received by gated PMT channel, and is present whenever there is a cloud along the beam path. This channel is used to measure

the distance to the cloud.

Solar blind PMT channel receives fluorescence signal, and is activated only when there is a suspicious cloud. Spectrograph with gated ICCD identifies nature of the bio-molecule responsible for fluorescence. Bandwidth of the receiver channel is usually kept smaller to disallow unwanted background radiation from entering the channel.

Most information on development of systems for stand-off detection of harmful bio-aerosols such as anthrax is classified. Some experimental systems are reported in public domain. One such system is UV LIF lidar system of Norwegian Defence Research Establishment (Fig. 6).

Fig. 7 shows a simplified schematic of Norwegian biological lidar. Early UV lidar systems used excitation at near 350nm. This wavelength had a fundamental problem. Many innocuous fluorophores, amino acids and other substances are also excited at that wavelength, making discrimination of threats challenging. Use of wavelength in 280nm region along with 350nm region enabled more effective weeding out of harmless targets and better identification of harmful bio-aerosols.

Another biological lidar system is the long-range biological stand-off detection system (LR BSDS). Shown in Fig. 8, it is used to detect, track and map large area aerosol clouds at ranges up to 30km. The system has three major components, including 100Hz operationally eye-safe laser transmitter, 61cm (24-inch) receiving telescope and transferred electron-intensified photo diode detector with an information processor integrated into a frame.

This system provides information about cloud configuration, such as size, shape, and relative intensity and cloud location including range, width, height, height above ground and drift

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